Name- Ayush Singh

Course Name – Data Analytics With GenAI

Subject – Excel

Topic – Data Cleaning & Functions

Contact Number – 7054891639

Email ID – as764994@gmail.com

7054891639

Question 1:

What is data cleaning, and why is it important in data analysis? What are the potential consequences of analyzing unclean or messy data? Explain the common steps involved in cleaning and organizing data.

Answer 1:

Data Cleaning: Data Cleaning means fixing the errors and properly organizing the messy and unstructured data and making it clean so that it can used to access some information and making it accurate also.

Important: If the data is not properly structured, we can extract useful and accurate information from it. Clean data gives correct and accurate information and leads to correct conclusion after making analysis.

Consequences of analyzing messy data:

1. Wrong business decisions
2. Incorrect reports
3. Wasted time and money

Common steps in data cleaning:

1. Remove duplicate entries
2. Fix spelling errors
3. Fill missing values
4. Remove irrelevant data
5. Standardize formats (like dates)

Example: If a customer's name is written as "Ayush," "AYUSH," and "ayush" in different rows, clean it to just "Ayush."

7054891639

Question 2:

How would you sort the following dataset first by "Department" (A-Z) and then by "Salary" (Largest to Smallest)? Write a step-by-step approach.

Employee	Department	Salary
Sonu	IT	4000
Pranav	HR	5000
Rahul	IT	2500

Answer 2:

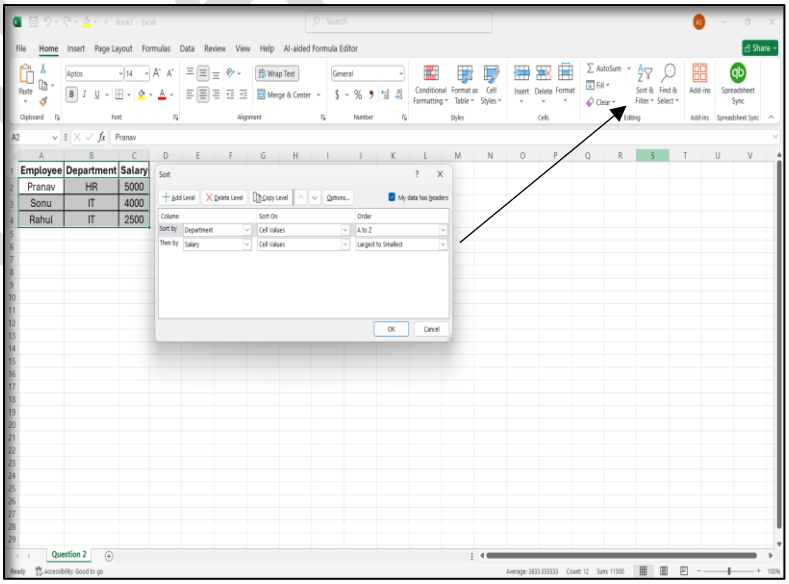
Sort by Department (A-Z), then by Salary (Largest to Smallest)

Steps to follow: (We will use custom sort)

1. First, sort the Department column alphabetically (A-Z)
2. Then, within each department, sort the Salary from highest to lowest

Employee	Department	Salary
Pranav	HR	5000
Sonu	IT	4000
Rahul	IT	2500

Here is the Screenshot of this:



Here is Link to my **Excel Workbook**

7014891639

Question 3:

Explain the use of text functions such as TRIM , LEFT, RIGHT, MID, and CONCAT in data cleaning.

Answer 3:

1. TRIM: Removes extra spaces from the text (from start, middle and end)

- Example: " Ayush Singh " – "Ayush" **[=TRIM(" Ayush Singh ")]**

2. LEFT: Extract the characters from the left side of text

- Example: " Ayush " – "Ayu" **[=LEFT("Ayush",3)]**

3. RIGHT: Extract the characters from the right side of text

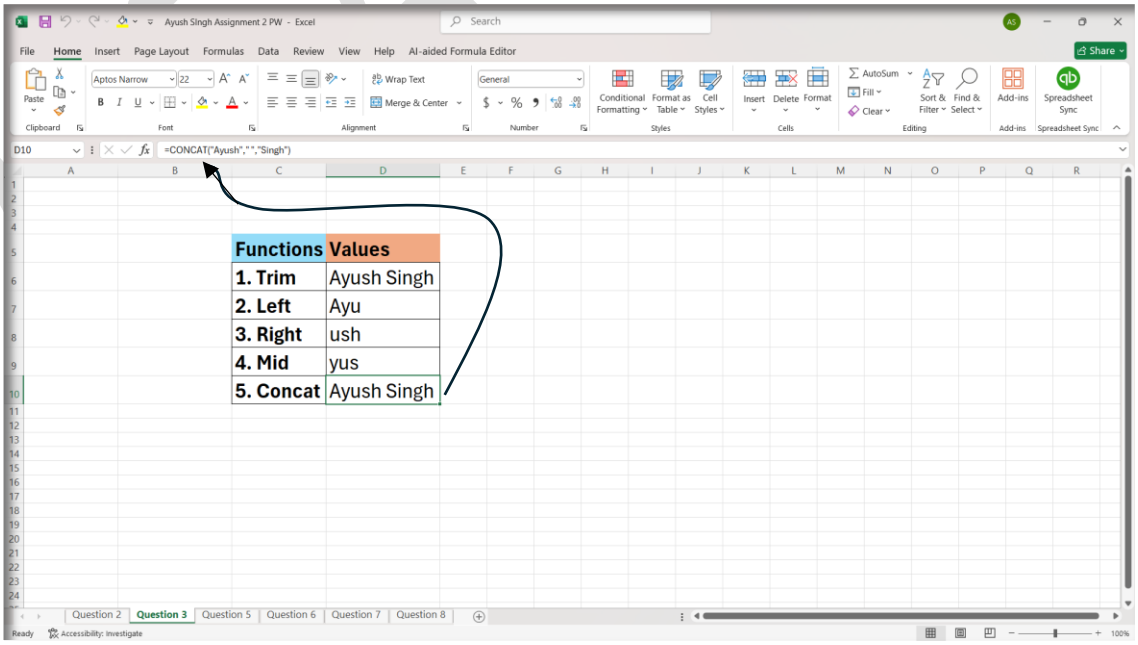
- Example: " Ayush " – "ush" **[=RIGHT("Ayush",3)]**

4. MID: Extract the characters from the middle of the text

- Example: " Ayush " – "ush" **[=MID("Ayush",2,3)]**

5. CONCAT: Joins 2 or more texts together

- Example: "Ayush" and "Singh" – "Ayush Singh" **[=CONCAT("Ayush"," ","Singh")]**



Functions	Values
1. Trim	Ayush Singh
2. Left	Ayu
3. Right	ush
4. Mid	yus
5. Concat	Ayush Singh

Here is Link to my **Excel Workbook**

7-489639

Note:

(In Excel, the **CONCAT** function supports **3 arguments**, so we can join more than two values easily.
But when I convert my Excel file into **Google Sheets**, it shows an error because Google Sheets **CONCAT only takes 2 arguments**.
That is why it shows **AyushSingh** instead of **Ayush Singh**, because the space becomes the third argument.
If we want a space, we should use **CONCATENATE** or **TEXTJOIN** in Google Sheets.
But since the question specifically asked to use **CONCAT**, I used that only.)

Question 4:

What is the role of date functions like TODAY in managing datasets?

Answer 4:

Role: Automatically shows today's date

Use: Track how old records are, calculate due dates, filter recent data

Example: If we'll write =**TODAY()** in function tab it will return **07/12/2025** and **order date is 01-Dec-2025**, you can calculate it's **6 days old**.

7054891639

Question 5:

Apply Data Validation to restrict Quantity values to only whole numbers between 1 and 10.

a) **Configure an input message that appears when a user selects a cell in the "Quantity" column explaining:**
   "Please enter a whole number between 1 and 10."

b) **Set up an error alert message that triggers if the user enters a number less than 1 or greater than 10, showing:**
   "Invalid input! The quantity must be a whole number between 1 and 10."

Answer 5:

Step-by-step approach:

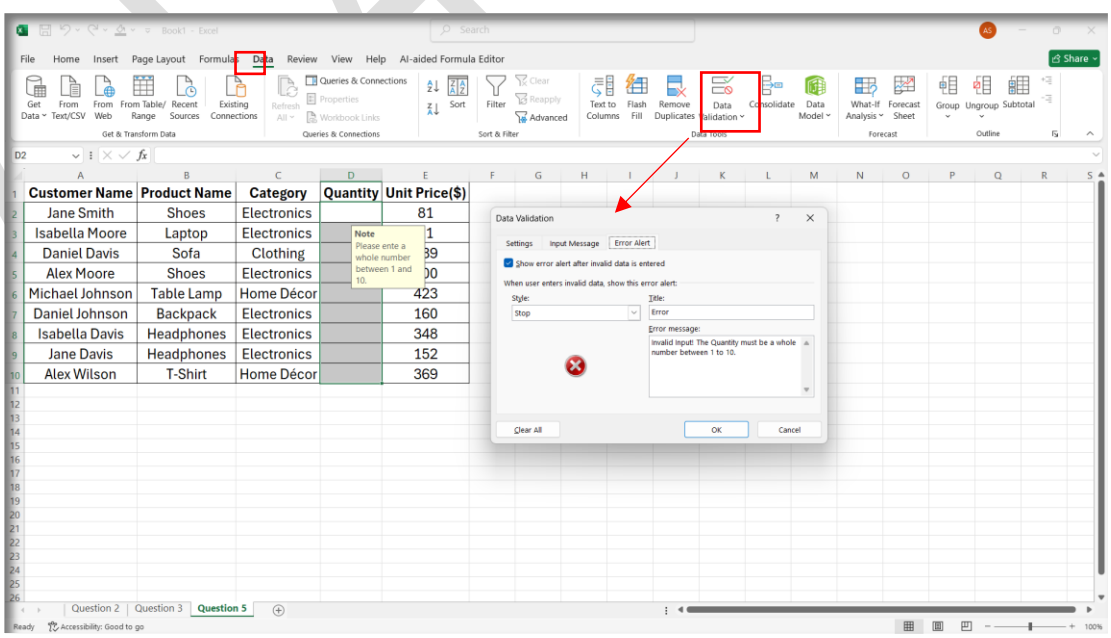
a. Input message:

1. Select the Quantity column
2. Go to **Data** → **Data Validation**
3. Choose "Whole Number" between 1 and 10
4. In "Input Message" tab, write: "Please enter a whole number between 1 and 10."

b. Error alert:

1. In "Error Alert" tab, choose "Stop" style
2. Write message: "Invalid Input! The quantity must be a whole number between 1 and 10."

Here is the Screenshot of this:



Here is Link to my **Excel Workbook**

Question 6:

Understand and apply fundamental text functions like LEFT, RIGHT, MID, and LEN.

- **Extract the first 5 characters from the string "ExcelTipsAreGreat" using the LEFT function.**
- **Extract the last 4 characters from "DataAnalysis.xlsx" using the RIGHT function.**
- **Extract the substring "Tips" from "ExcelTipsAreGreat" using the MID function.**
- **Count the total number of characters in the string "Hello World!" using the LEN function.**
- **Create a formula to extract the middle 6 characters from "12345-67890-ABCDE".**

Answer 6:

a. Extract first 5 characters from "ExcelTipsAreGreat":

- Formula: **=LEFT("ExcelTipsAreGreat", 5)**
- Result: **Excel**

b. Extract last 4 characters from "DataAnalysis.xlsx":

- Formula: **=RIGHT("DataAnalysis.xlsx", 4)**
- Result: **xlsx**

c. Extract "Tips" from "ExcelTipsAreGreat":

- Formula: **=MID("ExcelTipsAreGreat", 6, 4)**
- Result: **Tips** (starts at position 6, takes 4 characters)

d. Count characters in "Hello World!":

- Formula**: =LEN("Hello World!")**
- Result: **12** (including space and exclamation mark)

e. Extract middle 6 characters from "12345-67890-ABCDE":

- Formula: **=MID("12345-67890-ABCDE", 7, 6)**
- Result: **67890-** (starts at position 7, takes 6 characters)

Here is Link to my **Excel Workbook**

70549 1639

Question 7:

Understand how to combine text using CONCAT, TEXTJOIN, and the & operator.

a) **Use CONCAT to combine "Hello" and "World" with a space in between.**
b) **Combine "Apple", "Banana", and "Cherry" into a single string separated by commas using TEXTJOIN.**
c) **Use the & operator to create the string "2025: Excel Functions" by combining "2025", ": ", and "Excel Functions".**
d) **Create a comma-separated list from the range A1:A5 using TEXTJOIN**
e) **Combine first names in column A with last names in column B to create full names in column C.**

Answer 7:

a. Use CONCAT to combine "Hello" and "World" with space:

- Formula: **=CONCAT("Hello", " ", "World")**
- Result: **Hello World**

Note:

(In Excel, the **CONCAT** function supports **3 arguments**, so we can join more than two values easily.
But when I convert my Excel file into **Google Sheets**, it shows an error because Google Sheets **CONCAT only takes 2 arguments**.
That is why it shows **HelloWorld** instead of **Hello World**, because the space becomes the third argument.
If we want a space, we should use **CONCATENATE** or **TEXTJOIN** in Google Sheets.
But since the question specifically asked to use **CONCAT**, I used that only.)

b. Combine "Apple", "Banana", and "Cherry" with commas using TEXTJOIN:

- Formula: **=TEXTJOIN(", ", TRUE, "Apple", "Banana", "Cherry")**
- Result: **Apple, Banana, Cherry**
- **Explanation: TEXTJOIN** adds **comma+space** between each word automatically

c. Use & operator to create "2025: Excel Functions":

- Formula: ="2025: " & "Excel " & "Functions"
- Result: **2025: Excel Functions**
- **Explanation:** The **&** symbol joins text pieces together

7054891639

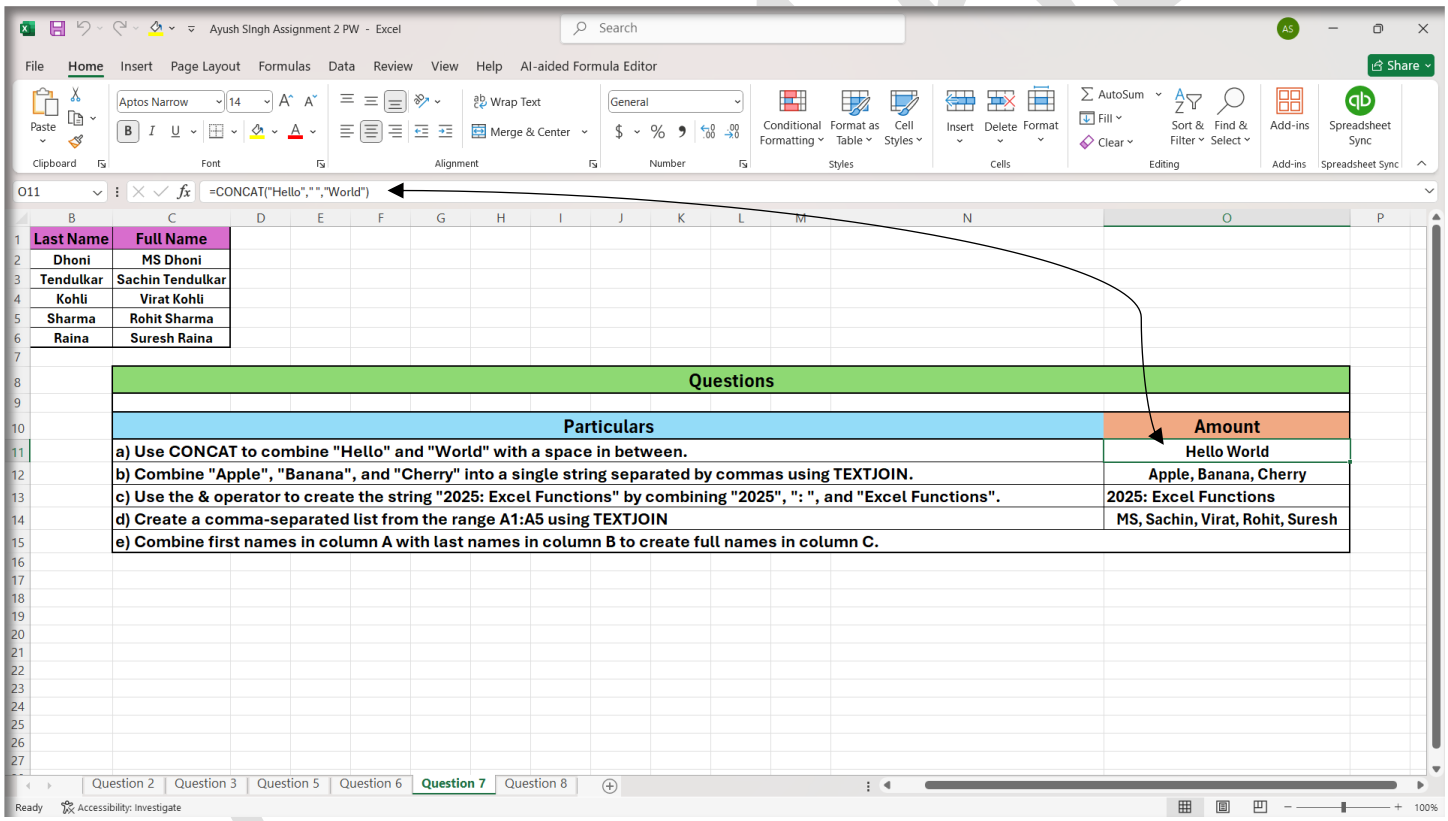
d. Create comma-separated list from range A1:A5 using TEXTJOIN:

- Formula: =TEXTJOIN(", ", TRUE, A1:A5)
- **Example:** If A1:A5 contains: Apple, Banana, Cherry, Mango, Grape
- Result: **Apple, Banana, Cherry, Mango, Grape**

e. Combine first names (Column A) with last names (Column B) to create full names in Column C:

- Formula in C1: =A1 & " " & B1
- **Example:**
  - A1 = "MS", B1 = "Dhoni"
  - C1 = **MS Dhoni**

Here is the Screenshot of this:



Here is Link to my **Excel Workbook**

Question 8:

Understanding TODAY() and NOW()

a. What is the difference between TODAY() and NOW() in Excel? Provide an example of when you would use each function.

b. If cell A1 contains the date 2025-06-10, write a formula using TODAY() to determine how many days are left until that date

c. Write an Excel formula using NOW() to display the current date and time in the format MM/DD/YYYY HH:MM AM/PM. ´

d. If a cell contains =TODAY(), what will happen when the worksheet is reopened the next day? Explain

e. You want to store a static date (today's date) in a cell without it changing every day. What keyboard shortcut should you use?

Answer 8:

a. Difference between TODAY() and NOW():

Function	What it shows?	Example
TODAY()	Only current date	12/7/2025
NOW()	Current date AND time	12/7/2025 3:46

When to use each:

- Use **TODAY()** when you only need the date (like calculating age, days until deadline)
- Use **NOW()** when you need exact timestamp (like recording when data was entered)

Note: In Excel my formula **=A1-TODAY()** gave **–180**, but in Google Sheets the same formula gave **–179** because both programs use slightly different date systems (Excel has an extra old leap-day in its calendar), so sometimes the day difference becomes off by one when I open the same file in Google Sheets.

b. Formula to calculate days left until 2025-06-10:

- Formula in any cell: **=DATE(2025,6,10) - TODAY()**
- Or if A1 contains the date: **=A1 - TODAY()**
- **Example:** If today is 07-Dec-2025, days left = -**180 days**

7054891639

c. Display current date and time in format MM/DD/YYYY HH:MM AM/PM:

- Formula: **=TEXT(NOW(), "MM/DD/YYYY HH:MM AM/PM")**

Result: **12/07/2025 05:08 AM**

d. What happens when worksheet is reopened if cell contains =TODAY()?

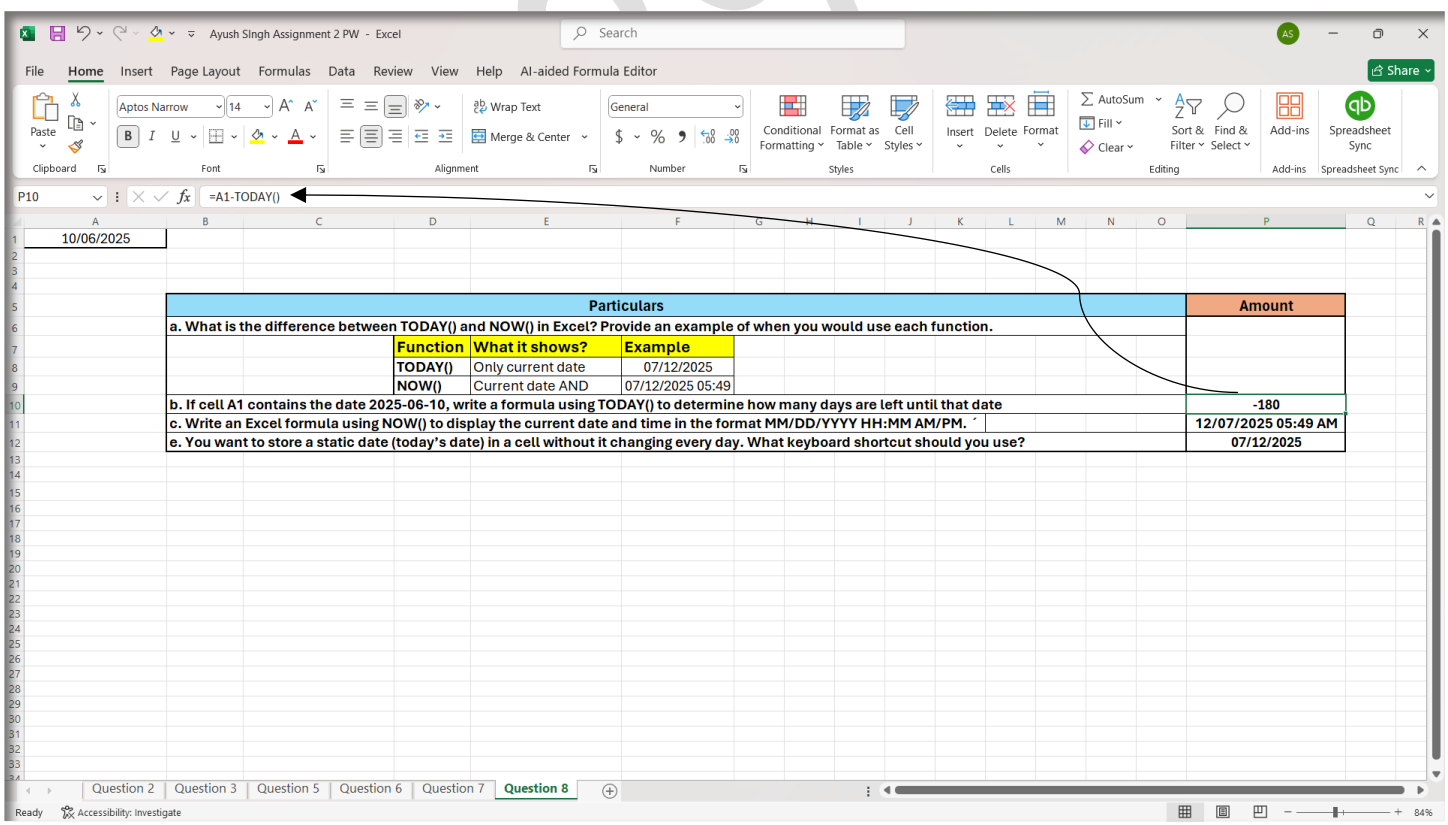
- **Answer:** The date will automatically update to the new current date
- **Explanation:** TODAY() is a dynamic function - it always shows the current date when the file is opened
- **Example:** Today it shows 07-Dec-2025, tomorrow it will show 08-Dec-2025

e. How to store a static date without it changing?

- **Keyboard shortcut:** Press **Ctrl + ;** (semicolon)
- This enters today's date as a fixed value, not a formula
- **Example:** If you press Ctrl+; today, it enters 06-Dec-2025 and will **NEVER** change, even tomorrow

Alternative: Type the date manually like 07/12/2025

Here is the Screenshot of this:



Here is Link to my [Excel Workbook]

7054891639